

# Problem Statement and Goals

## Squigglr

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Table 1: Revision History

| Date              | Developer(s)      | Change                                                                                                                                    |
|-------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 26 September 2026 | Rayan Nasrallah   | Added the Problem Statement section, covering the problem, inputs and outputs, stakeholders, and environment, with supporting references. |
| 26 September 2026 | Rayan Nasrallah   | Added the reflection responses to the appendix and the generative AI use disclosure.                                                      |
| 26 September 2026 | Moustafa Moustafa | Added the Goals and Stretch Goals section.                                                                                                |
| 26 September 2026 | Moustafa Moustafa | Added my reflection responses to appendix.                                                                                                |

## 1 Problem Statement

### 1.1 Problem

Visual-motor integration is the ability to coordinate visual information with movement. Children rely on it when drawing, copying shapes, and learning to write. Research links these skills to handwriting legibility and identifies motor or handwriting difficulties in some children with attention-deficit/hyperactivity disorder (ADHD) or autism [1–3]. The need extends from kindergarten and primary school learners to older children who require continued support.

Children need opportunities to practise at an appropriate level, understand their difficulties, and develop greater independence. Parents and educators need to distinguish what a child can do independently from what they accomplish with assistance. A finished drawing alone does not provide that distinction. Research on structured foundational activities supports the value of practice for visual-motor development [4].

There is an opportunity to build a system that supports drawing practice suited to individual abilities, provides understandable feedback, and helps children and supporting adults track independent progress. The intended outcome is improved drawing skills and greater independence, with opportunities for creative expression. Supporting adults need access to both individual attempt records and understandable summaries of change over time. Clinical diagnosis and treatment are outside the project scope.

## **1.2 Inputs and Outputs**

### **1.2.1 Inputs**

- Selected drawing activities and support preferences.
- Children’s drawing attempts during lessons and creative activities.
- Editing actions, such as moving, resizing, recolouring, or erasing drawing elements.

### **1.2.2 Outputs**

- Drawing targets, demonstrations, and guidance appropriate to the activity and support level.
- Simple activity feedback and completion results based on predefined observations.
- Editable creative artwork, including supported automatic refinements and recoverable original drawings.
- Attempt records showing encountered difficulties, changes within an attempt, guidance used, and the final outcome.
- Progress summaries showing changes across attempts and recurring difficulties, with access to the supporting records.

## **1.3 Stakeholders**

- Kindergarten and primary school children.
- Children of different ages who need additional support, including children with autism, ADHD, or other developmental or learning differences.
- Parents, guardians, and caregivers.

- Educators and learning-support staff.
- School administrators.
- Domain advisers in inclusive education and accessible educational software.
- Dr. Christopher Anand, project supervisor.

## **1.4 Environment**

Users will need access to a supported web browser on a compatible device with a display and a supported drawing input. Using the application will not require installing a separate application.

# **2 Goals**

## **2.1 G1. Provide structured practice in lines, shapes, and simple drawings**

Provide structured opportunities to practice lines, shapes, and simple drawings that involve visual-motor coordination. This goal is achieved when users can complete practice activities covering the agreed foundational skills, receive feedback, and repeat an activity. The team will verify this by completing each activity in the lesson catalogue.

## **2.2 G2. Enable progression from tracing to drawing without guides**

Enable learners to progress from tracing to drawing with reduced guidance and eventually without guides. This goal is achieved when the application supports this progression and restores assistance following repeated difficulty. The team will verify both directions using prepared sequences of successful and unsuccessful drawing attempts.

## **2.3 G3. Adapt drawing activities to individual guidance and difficulty needs**

Allow learners with different starting abilities and support needs, including children with autism or ADHD, to access suitable practice. This goal is achieved when the same activity can be completed with different guidance and difficulty settings. The team will verify that changing these settings produces the intended differences in the activity.

## **2.4 G4. Support free drawing with automatic refinement**

Allow children to draw freely beyond structured lessons, with automatic assistance for supported shapes and simple objects. This goal is achieved when users can create drawings of their choosing, receive automatic refinement, and continue editing the result while retaining the ability to restore their unmodified drawing. The team will verify this by creating and refining supported drawings in the creative space, then modifying and undoing the corrections.

## **2.5 G5. Record activity attempts and encountered difficulties**

Help parents and educators understand a child's progress by recording each activity attempt, the guidance provided, difficulties encountered, and the final outcome. Difficulties will be captured through predefined observations appropriate to the activity, such as missed tracing points or incomplete lines. This goal is achieved when attempts can be reviewed and compared without losing earlier difficulties or overlooking improvements made within an attempt. The team will verify this using prepared attempts with known difficulties and outcomes, confirming that the stored records and progress summaries accurately reflect them.

## **2.6 G6. Help parents and guardians understand their child's progress and practice needs**

Allow parents and guardians to review their child's activity history, recognise improvements, and identify recurring difficulties that may warrant further practice or discussion with an educator. This goal is achieved when an authorised adult can access a summary of completed activities, changes in performance, and assistance used, and inspect the attempts supporting that summary. The team will verify this using prepared histories showing improvement, repeated difficulties, and changing support levels, checking that each summary accurately represents the underlying records.

# **3 Stretch Goals**

**Camera-based finger tracing.** Display shapes, tracing guides, and start and end points over a live camera view. Children would see themselves in the background and trace the guides by moving their finger, offering another fun way to practise visually guided movement.

**Letter tracing and drawing.** Extend the activities to letters, allowing children to progress from tracing letter outlines to forming letters without guides.

## 4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

## Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

**Rayan:** Through my parents, I connected with a specialist who holds a doctorate and develops software for children with autism, as well as the head of a nonprofit organisation serving people with additional support needs, including autism and ADHD. These connections gave us a clearer direction for identifying stakeholders who could inform our requirements. They also helped me recognise the value of seeking both software-design expertise and experience supporting our intended users. We now have potential contacts who can help us question our assumptions and understand needs that we might otherwise overlook.

**Ahmed Elzaria:** What went well was developing a clearer connection between our users' needs and the goals of the application. I found the distinction between completing a drawing and completing it independently particularly useful. A finished drawing can look successful without showing how much guidance the child needed. Recording assistance and encountered difficulties therefore gives the progress summaries a clearer purpose than simply displaying completed artwork.

Working on the Development Plan alongside this document also helped me consider whether our proposed approach supported the stated goals. For example, automatic refinement should leave children with artwork they can continue editing, which affects what we need to investigate in the proof of concept. I learned that dividing the documents between team members does not remove the need to check their consistency. For future milestones, I would make this comparison an explicit review step so that our technical plans continue to address the agreed user needs.

**Moustafa:** After the first tutorial in the second week, Rayan and I discussed two project ideas with our TA, Chris Schankula. Chris provided valuable insights into what we should consider and how we could approach the drawing application. He also recommended Dr. Anand as a potential

supervisor. This discussion helped us evaluate the idea more clearly and gave us the confidence to pursue it. Having an early opportunity to discuss the project with someone knowledgeable helped establish a direction before we committed further effort to the proposal.

2. What pain points did you experience during this deliverable, and how did you resolve them?

**Rayan:** When I first presented our idea to Dr. Smith, he explained that we needed to justify the need for the application. I realised that describing useful features was not enough; we needed evidence supporting the problem we wanted to address. I reviewed research on visual-motor development, drawing practice, and difficulties experienced by some children with autism or ADHD. Comparing these findings helped me connect our proposal to existing work and explain the rationale for structured drawing practice. It also clarified the distinction between evidence supporting our approach and evidence that our particular application will be effective. In future deliverables, I would establish the evidence and its limitations earlier so that they guide our scope from the beginning.

**Ahmed Elzaria:** A challenge for me was separating an interesting technical feature from a convincing reason to build it. Our earlier idea included drawing in the air using a camera, but the professor's feedback challenged whether that interaction addressed a meaningful need. This made me reconsider how easily enthusiasm for an implementation can influence the way we describe the problem.

We responded by focusing the core experience on touchscreen or stylus drawing and keeping camera-based interaction as a stretch goal. Our supervisor's recommendation to develop a web application also gave us a more focused implementation direction. However, choosing a practical platform does not establish that the application will benefit its users. I see the research supporting the problem and future stakeholder feedback as necessary complements to technical feasibility.

Another point that needed clarification was what automatic refinement would provide. Returning a separate enhanced image would not meet our intention for children to continue editing their drawings. Clarifying this distinction helped connect the goal to an observable result. In future, I would describe what a user must be able to do after each proposed feature before discussing the technology used to build it.

**Moustafa:** We initially struggled to write goals at the right level. After discussing them with Chris, we concluded that they were too broad, and he emphasised that they needed to be measurable. Our first response was to add numerical targets to the existing goals, but this felt forced and introduced evaluation expectations that depended on testing with children. We then reconsidered what users should actually be able to accomplish and wrote new goals with observable completion criteria that

our team could verify. This helped me understand that a measurable goal does not necessarily require a number to be quantifiable; it needs a clear way to establish whether the intended outcome has been delivered. I would apply that distinction earlier when defining goals for future work.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

**Team response:** We narrowed the core scope to structured drawing practice, progression towards independent drawing, adjustable support, free drawing with refinement, attempt recording, and parent-facing progress summaries. We moved camera-based tracing and letter activities to stretch goals so that they would not distract from completing the core experience. We also limited activity feedback to predefined observations, such as missed tracing points, rather than personalised generated advice.

We distinguished user goals from supporting requirements such as account creation and browser compatibility. We also separated recording individual attempts from summarising progress, giving each goal a distinct purpose. Our completion checks now use prepared drawing attempts and activity histories that the team can evaluate without depending on child participants. These decisions keep the project manageable while retaining substantial design work in drawing assessment, adaptive guidance, editable artwork, and accurate progress reporting.

## Generative AI Use Disclosure

Our team used OpenAI GPT-6 Astra to improve the presentation of this document. We independently developed the project ideas and wrote the initial content before using AI. Its role was limited to L<sup>A</sup>T<sub>E</sub>X formatting, citation formatting, grammar correction, and improvements to sentence structure, clarity, and flow. No additional tools or skills were used.

All project ideas and substantive decisions originated with the team. We reviewed the suggested edits against our original draft to ensure that they preserved our intended meaning and checked the formatted citations against the source materials.

## References

1. Daly, C. J., Kelley, G. T., and Krauss, A. (2003). Relationship between visual-motor integration and handwriting skills of children in kindergarten: A modified replication study. *American Journal of Occupational Therapy*, 57(4), 459–462. <https://doi.org/10.5014/ajot.57.4.459>
2. Kaiser, M.-L., Schoemaker, M. M., Albaret, J.-M., and Geuze, R. H. (2015). What is the evidence of impaired motor skills and motor control among children with attention deficit hyperactivity disorder (ADHD)? Systematic review of the literature. *Research in Developmental Disabilities*, 36, 338–357. <https://doi.org/10.1016/j.ridd.2014.09.023>
3. Fuentes, C. T., Mostofsky, S. H., and Bastian, A. J. (2009). Children with autism show specific handwriting impairments. *Neurology*, 73(19), 1532–1537. <https://doi.org/10.1212/WNL.0b013e3181c0d48c>
4. Taverna, L., Tremolada, M., Dozza, L., Zanin Scaratti, R., Domahs, U., Lallo, C., and Toso, B. (2020). Who benefits from an intervention program on foundational skills for handwriting addressed to kindergarten children and first graders? *International Journal of Environmental Research and Public Health*, 17(6), 2166. <https://doi.org/10.3390/ijerph17062166>